Samples, Studies, and Bias KEY

Name _____

A radio station asks listeners to call in: *Should school start later?* 90% of callers say yes. Does that mean 90% of all students agree? Not necessarily.

	Meaning	**Example**
Population	The **whole group** we want to learn about.	All 1,200 students at a school.
Sample	The part of the population we collect data from.	100 of those students.

We can only **generalize** (say the results are true for the population) if the sample represents the population well.

Three types of studies

Sample survey	Ask a sample questions.	Ask 100 students how many hours they sleep.
Observational study	Watch or measure. Do **not** change anything.	Compare the grades of students who do and do not play sports.
Experiment	**Do something** to the subjects (a treatment) and measure what happens.	Give half the students a 10-minute break before a test. Compare scores.

Bias is when the way we collect data favors some answers. A biased sample does not represent the population.

Type of bias	**What happens**	**Example**
Voluntary response	People choose if they answer. People with strong feelings answer most.	The radio call-in poll.
Convenience	The sample is whoever is easy to reach.	Asking only your friends.
Undercoverage	Some groups are left out of the sample.	Calling only home phones (many people do not have one).
Nonresponse	People who were chosen do not answer.	Half of the mailed surveys never come back.
Response	The question or the setting pushes people to answer a certain way.	“Don’t you agree that our great principal deserves a raise?”

Work with a partner. Compare answers when you finish each problem.

1. Name the population and the sample.

   (a) A city wants to know how residents feel about a new park. It surveys 400 residents.
   Population: all residents of the city. Sample: the 400 residents surveyed.

   (b) A farmer tests 50 apples from her orchard to check for worms.
   Population: all apples in the orchard. Sample: the 50 apples tested.

   (c) A school picks 30 seniors to ask about graduation plans.
   Population: all seniors at the school. Sample: the 30 seniors.

2. Is each study a sample survey, an observational study, or an experiment? Explain.

   (a) Researchers ask 500 adults how many hours of TV they watch.
   Sample survey. They ask questions and change nothing.

   (b) A coach has half the team stretch before practice and half not stretch. She counts injuries.
   Experiment. The coach assigns a treatment (stretching).

   (c) A scientist records the weights of babies whose mothers smoked and babies whose mothers did not.
   Observational study. The scientist does not choose who smokes.

   (d) Students give one plant fertilizer and another plant only water. They measure growth.
   Experiment. The students apply a treatment (fertilizer).

3. Name the type of bias. Explain how it could change the results.

   (a) A website asks visitors to vote on their favorite phone.

   Voluntary response. People who care a lot choose to vote, so the results may not match all phone users.

   (b) To learn about cafeteria food, a student asks people standing in the lunch line.

   Convenience (and undercoverage). Students who skip the cafeteria are left out, so the results may be too positive.

   (c) A survey about internet use is done only on the internet.

   Undercoverage. People without internet cannot be in the sample, so internet use will look higher.

   (d) A survey asks, “Do you agree that dangerous skateboarding should be banned from parks?”

   Response bias. The word “dangerous” pushes people to say yes.

4. To find the favorite sport of students at a school, Jamal surveys 60 fans at a basketball game. Can he generalize his results to all students at the school? Explain.

   No. Fans at a basketball game probably like basketball more than other students. The sample does not represent the whole school.

5. Rewrite this question so it does not push people toward an answer:

   “Don’t you think the school day is too long?”

   “Do you think the school day is too long, too short, or about right?”